

Industrial Component Anomaly Detection

Project Presentation

Unsupervised defect classification and pixel-level anomaly localisation from normal-only manufacturing parts.

Unsupervised AD

Deep Transfer Learning

Zero-Leakage Protocol

TEAM

B. Förster · A. Abul Hawa · M. Nadaf · K. Khalifa

MVTEC AD 2D Dataset

Selected Examples · Objects & Textures

CATEGORY

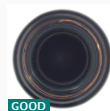
CONFORMING

DEFECTIVE

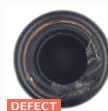
Bottle

Broken rim

GOOD



DEFECT



Capsule

Scratch

GOOD



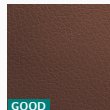
DEFECT



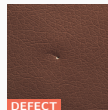
Leather

Surface cut

GOOD



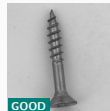
DEFECT



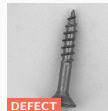
Screw

Thread defect

GOOD



DEFECT



Data Exploration

The MVTec AD Dataset – Class Imbalance

Dataset at a glance

- 15 manufacturing categories: 10 objects + 5 textures
- > 5000 images, $\approx$ 1200 anomalous images
- Normal-only training; test set has pixel-level anomaly ground truth

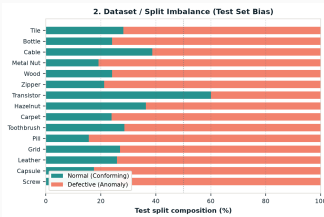
Accuracy paradox

- Median anomalous-pixel area: **1.54%**
- Predicting *all-normal* yields >98.5% pixel accuracy while catching zero defects.

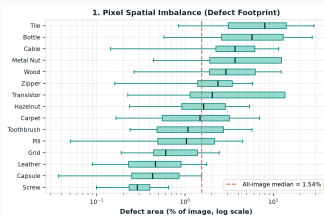
Kruskal-Wallis: $H(14) \approx 626.5$, $p < 0.001$

- $43\times$ spread in defect size across categories.

TEST-SET CLASS IMBALANCE (images)



PIXEL-AREA IMBALANCE (defect region %)



Defect Taxonomy & Spatial Distribution

Two fundamentally different defect regimes:

Structural Objects (e.g. Screw, Cable)

- Defects cluster at specific locations
- Strong positional bias (shortcut risk)
- Foreground/background separation critical

Continuous Textures (e.g. Grid, Leather)

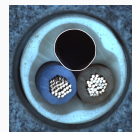
- Defects broadly distributed across surface
- Rich augmentation needed (rotation, scale)
- No foreground mask required

Covariate shift: Screw, Leather, Grid show measurable brightness/contrast drift between train and test normal images – motivates photometric augmentation.

OBJECTS – LOCALISED DEFECTS (contour = defect boundary)



Screw thread

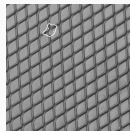


Cable missing

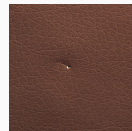


Capsule scratch

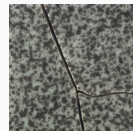
TEXTURES – DISTRIBUTED DEFECTS (defect across full surface)



Grid glue strip



Leather cut



Tile crack

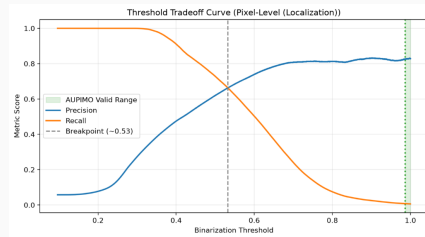
Metric Selection: AUPIMO & F1-Score

The Problem with Pixel-AUROC

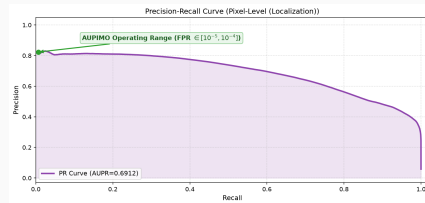
- AUROC relies on False Positive Rate (FPR):
 - With $\approx 98.5\%$ normal pixels, TN dominates $\Rightarrow$ FPR barely moves.
 - SOTA scores all exceed 99%, making differences indistinguishable.

Metric Solution

- **AUPIMO:** Integrates localization over the operational ultra-low FPR window $[10^{-5}, 10^{-4}]$ on normal images.
- **Dual Thresholds:** We calibrate distinct *image-level* and *pixel-level* anomaly thresholds.
- **F1-Score / PR Curve:** Directly measures defect capture vs false alarms without TN dilution.



Threshold tradeoff curve & AUPIMO range



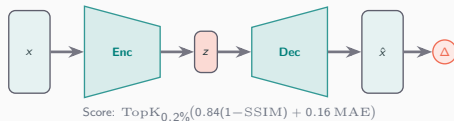
Precision-Recall curve with marked AUPIMO operating range

Models

Model Selection & Architecture Overview

1. Convolutional-Autoencoders

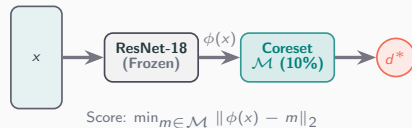
Paradigm: Generative Reconstruction · Built from scratch



- **Complete control:** Full freedom over latent bottleneck dimensions, skip connections, and loss constraints.
- **Intuition:** Trained on normal images; unable to reconstruct defects.
- **Trade-offs:** Prone to blurriness or identity shortcuts; sensitive to reconstruction thresholds.

2. PatchCore + ResNet-18 (Anomalib)

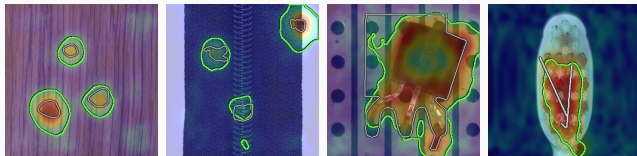
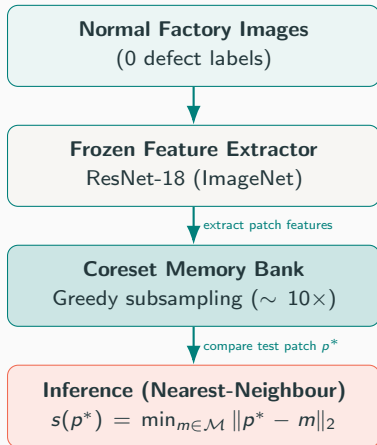
Paradigm: Deep Feature Memory Bank · High accessibility



- **High accessibility:** Zero model training required; out-of-the-box integration via Intel's `anomalib`.
- **ResNet-18 backbone:** Compact ImageNet representations; mid-level layers capture both texture and shape.
- **Coreset sampling:** Compresses the memory bank by 90%+ without losing local anomaly detection accuracy.

PatchCore: Transfer Learning & Architecture

Transfer Learning Paradigm



Wood, zipper, transistor, toothbrush. White = GT; green = predicted mask (pixel $q_{0.99}$).

Key Features

- **Cold-start eliminated:** Deployment without defect collection.
- **IP Protection:** Memory bank contains embeddings, not raw model weights.

Why it excels

- Mid-level layers capture local textures perfectly.
- No catastrophic forgetting (frozen backbone).
- ResNet-18 guarantees predictable GPU memory bounds.

Keras CAE – Autoencoder Learning Process

The Concept

Training Phase:

- Trained *excl.* on defect-free images.
- Learns the structural grammar of "normal" surfaces.

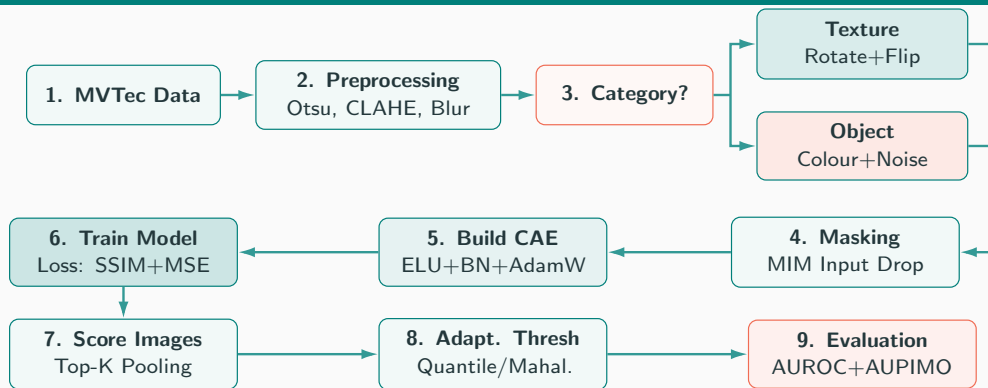
Inference Phase:

- On defective images, anomalies are reconstructed as "normal" parts.
- Reconstruction fails at the defect sites $\Rightarrow$ high reconstruction error.
- High reconstruction error $\Rightarrow$ Anomaly Signal.

Key Design Decisions

- **Masked Image Modelling (MIM):**
25% of patches randomly masked before encoding. Forces the model to understand global context, prevents "identity mapping".
- **SSIM + MSE Loss:**
 $\mathcal{L} = \alpha (1 - \text{SSIM}) + (1 - \alpha) \text{MSE}$.
MSE captures global variance, while SSIM enforces high structural and perceptual fidelity.
- **Architectural Choices:**
ELU no dying neurons + zero-centered mean.
BatchNorm stabilizes covariate shifts.
AdamW enables decoupled weight decay.

Keras CAE – End-to-End Pipeline



Automated Pipeline & Optuna Integration

The pipeline is dynamically orchestrated. Hyperparameters (Preprocessing/Model) (e.g., augmentation strategy, latent dimensions) optimized using Optuna targeting **Pixel AUPIMO**.

Evaluation Protocol & Benchmarks

Zero-Leakage Evaluation Protocol

The Core Principle

Every evaluation result in this presentation is produced under a deterministic, audit-compliant split that keeps the test set completely unseen until final scoring $\Rightarrow$ **zero-test-leakage**.

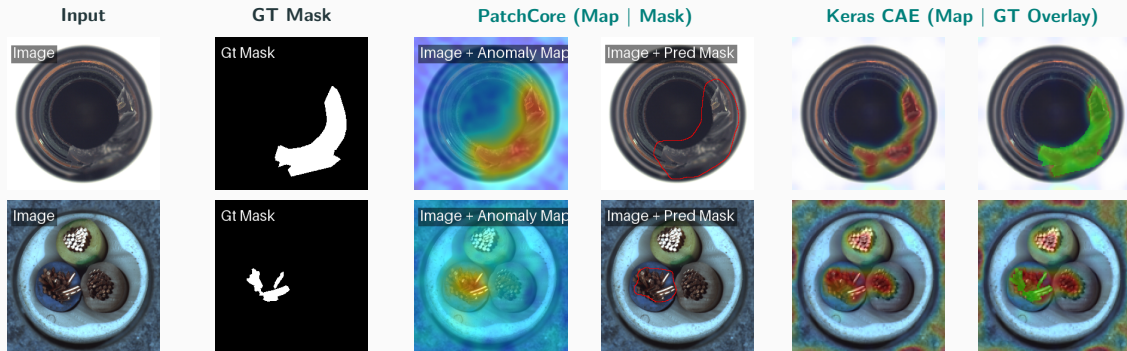
Peak Performance (PatchCore vs CAE)

Category	Metric	PatchCore	Keras CAE
Bottle (Object)	Image F1	0.962	0.992
	AUPIMO	0.982	0.507
Macro Average	Image F1	0.929	0.495
	AUPIMO	0.603	0.058

Canonical Evaluation Settings

- **Split Size:** 85% fitting / 15% validation
- **Test Set:** held out - normal + anomalies
- **Resolution:** 256×256 pixels.
 - Bilinear interpolation for continuous heatmaps.
 - Nearest-neighbor + binarization for GT masks.
- **Threshold:** 95th percentile ($\alpha = 0.05$ false-alarm budget).
- **AUPIMO:** Standard bounds
 $\text{FPR} \in [10^{-5}, 10^{-4}]$, 50k thresholds.

Qualitative Results - Localisation Comparison



Key Observations:

- **PatchCore:** Heatmaps sharply bound the defect, leading to precise predicted masks.
- **Keras CAE:** Reconstruction errors diffuse. While broadly highlighting anomalous regions, they fail at fine-grained pixel localization.

Business Interpretation

Economic Error Asymmetry — The Cost Model

Two types of error, very different costs

False Alarm (FP): good part rejected.

Cost: line stoppage, lost throughput, scrapped unit

e.g. *Screw*: $c_{FP} \gg c_{FN}$

Missed Defect (FN): bad part ships

Cost: recall, liability, safety incident

e.g. *Pill*: $c_{FN} \gg c_{FP}$

Optimal threshold $t^* = \frac{c_{FP}}{c_{FP} + c_{FN}}$

High c_{FN} (pills) pushes t^* toward zero - catch everything. High c_{FP} (screws) pushes t^* toward one - avoid false alarms.

Category	Obj.	t^*
Fasteners (Screw)	$F_{0.5}$	≈ 0.9999
Packaging (Bottle)	F_1	≈ 0.0003
Automotive (M. Nut)	F_1	$\approx 4 \times 10^{-6}$
Life Sci. (Pill)	F_2	$\approx 10^{-8}$

Cost profiles are illustrative estimates from composite industry case studies (for the sake of the project - guesses). They demonstrate the spectrum of liability, not precise financial figures.

Threshold Calibration & Risk-Tiered Operating Profiles

Calibration formula

$$\tau_{\alpha} = \text{Quantile}_{1-\alpha}(\{s_i \mid x_i \in \mathcal{D}_{\text{val},\text{norm}}\})$$

α : plant's tolerable false-alarm budget. We offer three pre-declared operating profiles:

T1 - Life-Safety ($\alpha \approx 0.10, F_2$)

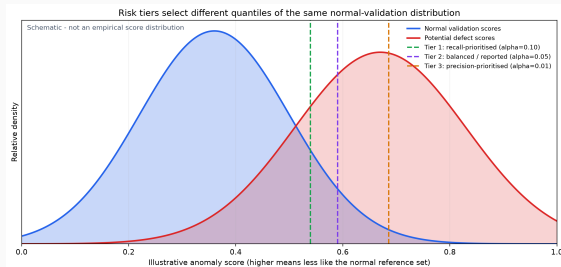
Recall weighted 4× over precision. Every borderline part is manually audited.

T2 - Prec. Industrial ($\alpha \approx 0.05, F_1$)

Balanced Type I / II errors. Reported benchmark.

T3 - High-Throughput ($\alpha \approx 0.01, F_{0.5}$)

Precision weighted 4×; prevents pseudo-scrap.



- **Schematic:** three profiles select different quantiles from the same normal-validation score distribution.
- **Key:** decoupling feature extraction from threshold calibration makes the decision gate a dynamic, auditable financial parameter.

SLA Interpretation & Operational Value

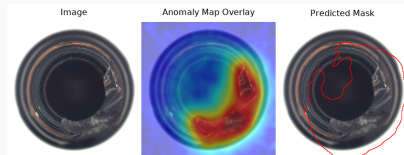
From academic metric to operational SLA

Metric	Operational meaning
Image F1 = 0.929	<8% of decisions are wrong
Recall = 0.932	$\leq 6.8\%$ of real defects missed
Precision = 0.936	$\leq 6.4\%$ of rejected parts are good
PR-AUC = 0.988	Model quality across all thresholds
AUPIMO = 0.603	Strong pixel-localisation

Day-1 Deployment & Cold-Start SLAs

No defect samples required at commissioning; onboarding uses only normal parts.

The α mechanism: W/o defect data, we cannot optimize F-scores. Day-1 SLA guarantees max false-alarm rate (α) on normal parts $\Rightarrow$ reliable proxy for target F-score.



Explainability: anomaly heatmaps provide pixel-level attribution
 $\Rightarrow$ quality engineers review in seconds, replacing manual inspection.

Data sovereignty

Coreset memory bank stores patterns locally. CAD toolpaths, material formulations never leave the plant.

Summary & Future Prospects

Results

- Unsupervised anomaly detection on MVTec AD 2D (15 categories)
- Two architectures: Keras CAE (generative) & PatchCore (transfer learning)
- Zero-leakage protocol; AUPIMO as primary pixel metric
- Business-grade threshold framework

Key design principles

- Cold-start: normal-only training, no defect labels
- IP protection: local coreset, no cloud upload
- Calibrated decision gate as financial parameter

Future Prospects & Experimental Work

- **Advanced CAE Architectures:**
 - **Targeted MIM:** Masked Image Modeling for object geometry, dual-weight masked loss computation.
 - **Perceptual Loss:** ResNet-18 feature-space penalization.
 - **Multi-scale SSIM:** Enhanced structural defect isolation.
- **Self-supervised ViTs (DINOv2/v3):** Foundation-model dense tokens breaking the convolutional representation ceiling.

Thank you

Questions welcome

B. Förster · A. Abul Hawa

M. Nadaf · K. Khalifa